

Synopsis

Optimal Route Planning in Logistics using A Search*

Problem Id: 11

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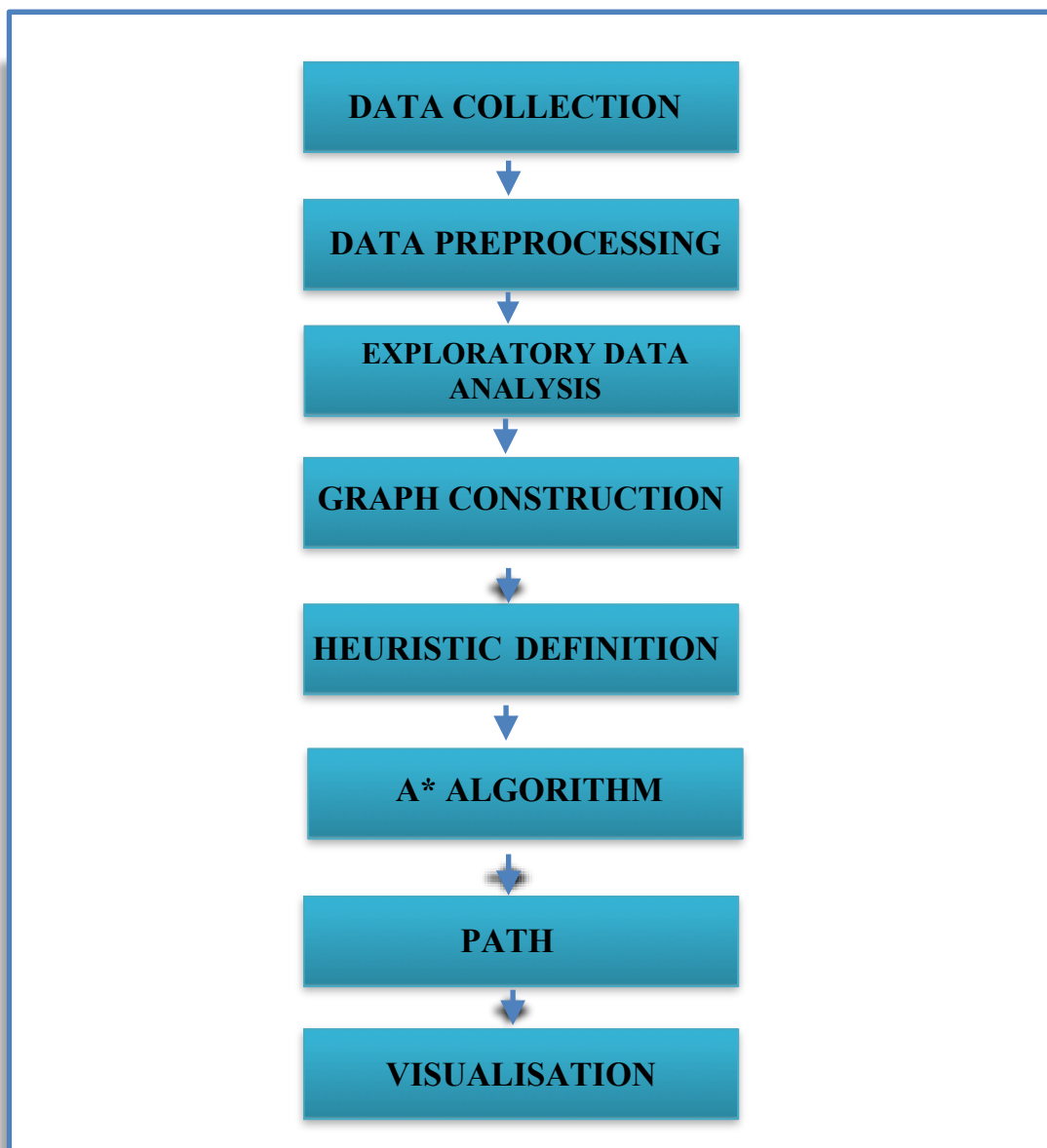
- **Problem Description:**

Logistic companies work on efficient route planning. If the company takes the wrong path or longer route, there it costs a lot of fuel and delays the delivery time. Finding the smallest path is important. However, selecting the best route from multiple possible paths can be challenging, especially when both actual distance and estimated proximity to the destination must be considered.

In this project we develop an intelligent logistics route planner using the A* (A-star) search algorithm. A weighted graph of 10 locations is designed, where nodes represent delivery points and edges represent road distances. A heuristic function is found using Euclidean distance formula. The A* algorithm combines actual cost and estimated cost to determine the most optimal path.

Example: It starts when a customer places an order on any e-commerce site, for the fastest delivery it requires the nearest warehouse to reduce delivery time, also while delivering there may be multiple possible routes, some shorter but congested, and others longer but faster. Therefore to complete this process algorithm like A* is used to find the optimal path efficiently so as to reduce any delays.

- **Steps of Implementation:**



- **Expected Modules/ Libraries to be used :**

- NumPy → Used for numerical computations and calculating Euclidean distance for the heuristic function.
- Pandas → Used to create and manage the dataset of nodes, edges, and distances in tabular form.
- Matplotlib → Used for visualizing the graph and highlighting the optimal path.
- NetworkX → Used to create and represent the weighted graph of locations and perform graph operations.
- Heapq → Used to create and represent the weighted graph of locations and perform graph operations.

- **Expected Evaluation metrics and Plots to be used:**

Evaluation Metrics

1. Path Cost (Total Distance) → Measures the total distance of the route found and determines whether the algorithm is giving the shortest and most efficient path.
2. Execution Time → Calculates how long the algorithm takes to find the path which is important for real-time logistics applications where quick decisions are needed.
3. Number of Nodes Explored → Indicates how many locations the algorithm checks before reaching the goal. Fewer nodes mean better efficiency.
4. Optimality → To ensure that the path found is actually the shortest.
5. Heuristic Accuracy → Ensures that the heuristic value is always less than or equal to the actual distance, which guarantees correct working of A*.

Visualization Plots

1. Graph/Network Plot → Shows all locations and connections and helps visualize how the logistics network is structured.
2. Shortest Path Highlight Plot → Displays the final route found by the algorithm, making it easy to understand the solution.
3. Distance Distribution (Histogram) → Shows how distances are spread across the dataset, ensuring the data is realistic.
4. Heuristic vs Actual Distance (Scatter Plot) → Used to verify that heuristic values are valid and do not exceed actual distances.
5. Performance Comparison Plot → Compares execution time or nodes explored for different algorithms (like A* vs Dijkstra), showing efficiency differences.

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- **Required topics from the subject:**

- File Handling → File handling allows programs to store and retrieve data from files. It helps keep data saved permanently.
- Pandas→Pandas is a Python library used for data analysis and manipulation. It helps to handle data in tables using structures like DataFrames and Series.
- Numpy→NumPy is a Python library used for numerical computations. It provides support for arrays and mathematical operations on large datasets efficiently.
- Outlier Detection→Outlier detection is the process of identifying data points that are significantly different from the rest of the dataset. It helps in improving data accuracy and analysis by removing or handling unusual values.
- Astar→A* (A-star) is a search algorithm used to find the shortest path between two points. It uses both actual cost and heuristic (estimated cost) to efficiently reach the goal.
- Heuristic→A heuristic is a method used to make quick decisions by estimating the best solution. It helps algorithms find answers faster without checking all possibilities.

- **Platform Required:**

- Jupyter Notebook
- Google Collab

- **Link Sources**

- Artificial Intelligence: A Modern Approach – for A* algorithm and heuristics
- Geopy Documentation – for distance calculation
- NetworkX – for graph visualization
- Matplotlib – for creating plots
- OpenStreetMap – for real location data